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Context-Aware Image Caption Editing via Hallucination-Resistant Visual Instruction Tuning

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Abstract

The goal of image caption editing is to revise a given caption so that it is semantically aligned with its corresponding image. Since it involves not only identifying inconsistencies in the caption but also correcting them based on the image, image caption editing remains a challenging problem. Previous approaches establish multimodal frameworks specifically designed for image caption editing. However, these methods do not leverage vision-language models (VLMs) built upon large language models (LLMs), which are expected to provide enhanced contextual understanding capabilities. In this paper, we propose leveraging recent VLMs for image caption editing. We observe that applying existing instruction-tuned VLMs often results in hallucinations for image caption editing. To address this issue, we introduce a new instruction tuning dataset which consists of two tasks for hallucination-resistant image caption editing. The first one is the multi-object selection task where the VLM is required to select the objects present in the image from a set of given objects, some of which include hallucination-inducing ones. The second task is the context-aware image caption editing task, where the model is required to modify a given image caption to align it with its corresponding image, while preserving the context of the original caption. Comparative results on two baseline VLMs demonstrate that our proposed instruction dataset leads to state-of-the-art performance in image caption editing. In addition, we empirically demonstrate that instruction tuning for context-aware image caption editing enhances robustness of VLMs against hallucinations.

1. Introduction

Image caption refers to a textual description that provides an explanation of the contents of an image. Since image captions contain detailed contextual information about the image, they are widely used to train various multimodal models that leverage both visual and textual data. Several representative studies that utilize image captions include automatic image caption generation [4, 22, 31, 44, 48], image retrieval using natural languages [16, 17, 35], and vision-language representation learning [25, 26, 39]. In many of these studies, the models are trained to learn the similarities between each image and its caption by attending to the major objects for visual-textual understanding. However, these models are not well suited for capturing the detailed discrepancies between the two modalities, as they focus primarily on the major contexts. A rigorous understanding of the detailed contexts of both modalities is crucial for developing reliable and explainable models.

Recently, image caption editing is addressed by several pioneering studies [41, 42, 49, 50]. Given a pair of image and reference caption, where a reference caption refers to a caption that contains inconsistencies with the image, the goal of caption editing is to generate a new caption which is semantically aligned with its corresponding image. Image caption editing is a challenging task since it involves not only identifying inconsistencies in the caption but also modifying them based on the image. Previous methods on image caption editing are categorized into implicit and explicit approaches. Implicit caption editing [41, 42] directly generates a new caption. Since implicit approaches allow rephrasing, it can result in distortions of the original structure and context of reference captions. Explicit caption editing [49, 50] applies a sequence of edit operations such as addition, deletion, and insertion to generate a new caption. In comparison to the implicit approaches, explicit approaches can produce more explainable and structure-preserving captions since it applies word-level edit operations to the reference caption. According to the recent studies [49], the explicit approaches lead to better editing performance than implicit methods. However, the existing methods do not leverage recent vision-language models (VLMs) which are expected to provide enhanced contextual understanding capabilities through large language models (LLMs) and multi-task visual instruction tuning. Rather, most of previous frameworks on image caption editing are developed based on relatively small-scale models such as the Up-Down at-

tention model [4] in [41, 42], ViLBERT [36] in [49], and Transformer [46] in [50].

Based on the above observations, we raise the question of whether adapting VLMs for image caption editing can lead to better performance. Through baseline experiments, we found that recent instruction-tuned VLMs often struggle with identifying and correcting inconsistencies between the reference caption and the image. To be specific, these models frequently produce hallucinated results or adhere too closely to the given reference caption while missing important details of the image contents. We hypothesize that the visual instruction datasets currently used to train VLMs lack of instructions that develop critical and analytical reasoning capabilities, which are essential for image caption editing. To address these issues, we introduce a visual instruction tuning dataset, developed through a principled methodology tailored for hallucination-resistant and context-aware image caption editing. The proposed dataset is composed of two kinds of visual instruction tuning tasks for training VLMs. The first one is the hallucination-resistant multi-object selection task, which requires the VLMs to select the object present in the image from a set of given objects. According to [28], VLMs are prone to generate hallucinations particularly for objects that frequently appear in visual instructions and those frequently co-occur with objects present in an image. Based on the finding, we propose a method to extract objects with a high potential to induce hallucinations for each image. Specifically, we propose to apply the association rule mining [2] to extract nonexistent yet frequently co-occurring objects in each image to obtain hallucination-inducing objects. By using both of the existent (i.e., ground-truth) objects and extracted nonexistent objects, we design instructions to require VLMs to select objects present in the image. This instruction enhances the perceptual capability and robustness to hallucination of the VLMs, making image caption editing results more reliable.

The second task in our proposed dataset is context-aware image caption editing, which requires the VLMs to revise a reference caption so that it is semantically consistent with the corresponding image while preserving its original context and structure. For dataset construction, we use GPT4o-mini [37] to generate reference captions for given pairs of image and its ground-truth caption. Inspired by the previous study [15], we introduce four aspects to generate reference captions, which are object existence, attribute, interaction, and count. This instruction facilitates the caption editing capability of VLMs by guiding them to identify and correct inconsistencies between the reference caption and the image. We provide experimental results on both domains of the existing explicit image caption editing [49] and the proposed context-aware image caption editing. We first show that applying VLMs for explicit image caption editing lead to better performance than the existing approaches but still not robust against hallucinations. Subsequently, we demonstrate that our proposed instruction tuning dataset leads to state-of-the-art results in context-aware image caption editing with enhanced perceptual capabilities. One of the key empirical observations of our study is the complementary relationship between context-aware image caption editing and hallucination robustness. To substantiate this finding, we present both quantitative and qualitative results of caption editing performance.

2. Related Work

2.1. Image Caption Editing

Early methods of image caption editing are proposed to refine captions generated by the image caption generation models [4, 22, 31, 44, 48]. The objective of image caption generation is to produce textual descriptions (i.e., captions) of given images. However, early methods of image captioning often struggle to capture fine-grained content details. To address this issue, Sammani *et al.* [41] propose a caption modification framework which can be plugged into existing caption generation models. By adding a residual gating module to the deep averaging network [21], the framework focuses on which parts to modify rather than generating captions from scratch. In [42], a selective copy memory attention mechanism is introduced, which captures the importance of words in the reference captions. These methods [41, 42] are the implicit approaches, which have a potential to distort the original structure and contexts of reference captions by the regeneration procedure.

To address the above limitation, Wang *et al.* propose an explicit caption editing framework called TIger [49]. It explicitly modifies reference captions by applying a sequence of edit operations, which are insertion, deletion, and addition operations, leading to more explainable and structure-preserving results than the implicit approaches. In [50], another explicit approach called DECap is introduced, which reformulates the caption editing task as a denoising process under the diffusion mechanism [5, 19]. However, these previous methods of image caption editing are based on the relatively small-scale models such as Up-Down attention [4] and ViLBERT [36]. We attempt to leverage recent VLMs that are trained using visual instruction tuning with LLMs. Our framework can be categorized to the implicit approach since the baseline VLMs in our work regenerate captions via autoregressive decoding given image-caption pairs. On the other hand, the objective of our proposed visual instruction tuning dataset is to generate structure-preserving and explainable captions similar to the explicit approach.

Table 1. Comparison of caption editing performance on COCO-EE and Flickr30K-EE datasets [49]. Metrics include BLEU-1 (B-1), BLEU-4 (B-4) [38], ROUGE-L (R) [29], CIDEr (C) [47], and SPICE (S)[3]. The best scores are shown in bold for each baseline VLM.

COCO-EE Test set					
Model	**B-1**	**B-4**	**R**	**C**	**S**
Ref-Caps	50.0	19.5	48.2	129.9	18.9
ETN [42]	53.8	23.8	53.3	190.5	32.1
TIger [49]	54.8	24.7	54.3	194.8	33.3
DECap [50]	56.1	23.4	53.1	177.0	32.2
LLaVA-1.5 [34] Baseline	50.2	20.0	49.5	140.3	22.5
+COCO-EE	**59.5**	**29.0**	**57.2**	**237.6**	**38.7**
mPLUG-Owl2 [53] Baseline	50.0	19.9	50.1	149.2	26.6
+COCO-EE	**58.5**	**27.9**	**56.5**	**231.0**	**38.2**

Flickr30K-EE Test set					
Model	**B-1**	**B-4**	**R**	**C**	**S**
Ref-Caps	34.7	10.9	36.9	91.3	23.4
ETN [42]	34.8	13.7	41.8	143.3	31.3
TIger [49]	38.3	14.9	42.7	148.3	32.0
DECap [50]	38.2	14.1	41.1	138.2	31.3
LLaVA-1.5 [34] Baseline	31.1	9.8	37.3	94.6	25.1
+Flickr30K-EE	**43.0**	**19.2**	**46.7**	**187.6**	**36.3**
mPLUG-Owl2 [53] Baseline	36.0	12.4	39.3	116.6	27.3
+Flickr30K-EE	**42.9**	**18.2**	**45.7**	**176.5**	**35.3**

2.2. Vision-Language Models (VLMs) and Hallucination in VLMs

The motivation behind early vision-language models (VLMs) stems from learning aligned feature representations of both textual and visual data. CLIP [39] is one of the pioneering models that learns feature representations through contrastive learning on both positive and negative image-text pairs. To extend VLMs to various applications, several multi-task architectures [25–27] are introduced. To leverage the advanced capabilities in natural language understanding, the methods for combining vision encoders and large language models (LLMs) [10, 11, 45] for VLMs are introduced by a variety of studies [8, 12, 33, 34, 52, 53, 56]. Especially, applying the instruction tuning mechanism [51] to VLMs, which is called visual instruction tuning, is the key approach to enhance the versatility of VLMs for handling various unseen tasks. Several representative VLMs that involve visual instruction tuning are LLaVA [33], LLaVA-1.5 [34], InstructBLIP [12], mPLUG-Owl1 [52] and mPLUG-Owl2 [53], MiniGPT4 [56], and MiniGPT-v2 [8]. Preparing reliable and high-quality visual instructions is essential to achieve the performance in various vision-language tasks.

Hallucination in VLMs refers to the generation of information that is not consistent with the contents of the given image, which severely degrades the reliability of VLMs. According to [28], VLMs are prone to generate hallucinations for visual objects which frequently appear or co-occur in the visual instruction datasets. In addition, most of the instruction-tuned VLMs tend to comply with negative instructions (i.e., invalid instructions such as requiring explanations about nonexistent objects) leading to hallucinations since they are trained with positive (i.e., valid) instructions only. To address these issues, various studies are introduced to relieve hallucination by introducing instruction datasets [7, 32], hallucination-resistant decoding [9, 14, 24], and training methodologies [55]. Since image caption editing involves rigid understanding on the given images, it is closely related to hallucination in VLMs. In this work, we focus on enhancing image caption editing ability by improving the robustness of VLMs against visual hallucinations using the proposed visual instruction tuning dataset.

3. A Visual Instruction Dataset for Context-Aware Caption Editing

In this section, we first introduce the motivation of establishing a new visual instruction tuning dataset, supported by our preliminary experimental results. Subsequently, we explain the details of the two instruction tasks of the proposed dataset.

3.1. Motivation from Preliminary Observations

To investigate the ability of VLMs in image caption editing, we conducted preliminary experiments by applying LLaVA-1.5 [34] and mPLUG-Owl2 [53] to the two test sets of the explicit image caption editing dataset released in [49], which are COCO-EE and Flickr30K-EE. As shown in Table 1, directly applying these baseline VLMs yields inferior results compared to previous methods. This is expected, as the vanilla VLMs are not trained for image caption editing tasks. Based on this observation, we construct visual instruction data for VLMs by leveraging the reference and ground-truth captions provided in the two train sets of the dataset. As demonstrated in Table 1, adding explicit image caption editing instructions leads to better performance than previous methods, which are specifically designed for the task. This observation supports our assumption that VLMs can be effectively leveraged to achieve state-of-the-art performance in image caption editing.

However, we observe that VLMs instruction-tuned on the explicit image caption editing task still tend to generate hallucinated outputs (see Fig. 3 and Table 3, discussed in Sec. 4). The explicit caption editing dataset [49] constructs reference captions by selecting negative samples based on image-caption similarity scores and edit distance. As a result, many of reference and ground-truth pairs lack strong semantic contrasts, which reduces the clarity and effectiveness of the editing task. In addition, robustness to hallucination may not be sufficiently improved, as the dataset does not explicitly consider hallucination-related cases. Moti-

vated by these observations, we propose a new instruction tuning dataset for image caption editing. The key objective of the proposed dataset is to enhance the ability of editing image captions by improving the perceptual capability of the VLMs. To this end, we establish two kinds of instruction tasks, which are described in the following subsections. For both tasks, we adopt sample lists from the previous work [49], which consist of 97K and 108K samples in the COCO [30] and Flickr30K [54] domains, respectively.

3.2. Instruction Task 1: Hallucination-Resistant Multi-Object Selection

The goal of the first instruction task is to enhance the perceptual robustness of VLMs against hallucination-inducing objects. To this end, we first extract nonexistent objects that have a high potential to cause hallucinations for each image. Subsequently, we establish a multi-object selection task, which requires VLMs to choose the objects that appear in each image, given an object list containing both existent and nonexistent objects. To obtain nonexistent and hallucination-inducing objects for each image, we propose leveraging the association rule mining scheme [2], which are widely used to discover relationships or patterns among a set of items in large datasets. Given a dataset D consisting of image-object pairs $D = \{(I_i, O_i)\}_{i=1}^{N}$, where each object set $O_i = \{o_1, o_2, \ldots, o_{N_i}\}$ represents the objects associated with image I_i, we aim to obtain the set of nonexistent objects $\widetilde{O}_i = \{\widetilde{o}_1, \widetilde{o}_2, \ldots, \widetilde{o}_{\widetilde{N}_i}\}$, where $i = 1, 2, \ldots, N$, with the potential to cause hallucinations. Following the association rule mining scheme, we define the support of an object set A_j as follows:

$$\text{Sup}(A_j) = \frac{|\{I_i : A_j \in O_i,\ i = 1, 2, \ldots, N\}|}{N}, \quad (1)$$

$$A_j = \{o_1, o_2, \ldots, o_{N_j}\}. \quad (2)$$

where N is the number of images in the dataset D, and the numerator in Eq. (1) represents the number of images whose object set O_i includes A_j. The support value represents the probability that the entire objects in A_j appear in an image in D. Since the number of possible object sets can be large, we apply the Apriori algorithm [1] to retain only those whose support exceeds a threshold $t_{\text{sup}} \in [0.0, 1.0]$. The set of frequent object sets is defined as follows:

$$S = \{(A_j, s_j) \mid j = 1, 2, \ldots, N_{\text{freq}}\}, \quad s_j = \text{Sup}(A_j). \quad (3)$$

In the above equation, N_{freq} is the number of frequent object sets whose support exceeds t_{sup}. Based on the frequently occurring object sets, our goal is to estimate the probability of co-occurring object sets for a given image. This objective can be quantified by the confidence between a pair of object sets, which is defined as follows [2]:

$$\text{Conf}(A_j \rightarrow A_k) = \frac{\text{Sup}(A_j \cup A_k)}{\text{Sup}(A_j)}, \quad \text{where} j \neq k. \quad (4)$$

Given an antecedent A_j, $\text{Conf}(A_j \rightarrow A_k)$ in Eq. (4) measures the probability of occurring the consequent A_k. Thus, the confidence represents the probability of co-occurring both object sets A_j and A_k for a given image containing A_j. In our task, object sets with high confidence are considered hallucination-inducing sets. We derive association rules from the frequent object sets by applying a confidence threshold $t_{\text{conf}} \in [0.0, 1.0]$, using MLxtend [40]. The set of association rules Z is defined as follows:

$$Z = \{(A_j, A_k, c_l) \mid j, k \in \{1, 2, \ldots N_{\text{freq}}\}, j \neq k, \\ l \in \{1, 2, \ldots N_{\text{rule}}\}\}, \quad (5)$$

where c_l and N_{rule} in Eq. (5) are $\text{Conf}(A_j \rightarrow A_k)$ and the number of association rules, respectively. Given the association rule Z, the confidence of consequent A_k (i.e., hallucination-inducing objects) can be estimated for a given antecedent A_j (i.e., objects occurred in the image). To decide the candidates of nonexistent objects, we obtain consequents of all possible antecedents (i.e., all subsets) of each object set O_i. Based on the consequents, we extract objects which are nonexistent in each image. Finally, the nonexistent object set $\widetilde{O}_i = \{\widetilde{o}_1, \ldots, \widetilde{o}_{\widetilde{N}_i}\}$ is obtained for each of the i-th image. To obtain the association rules, we set both t_{sup} and t_{conf} to 10^{-3} for the COCO domain and 5×10^{-4} for the Flickr30K domain.

The framework of generating visual instructions for the first task is illustrated in the upper part of Fig. 1. Given each image I_i and its corresponding existent object set O_i, we extract a set of nonexistent objects $\widetilde{O}_i$ using the proposed method. To introduce diversity and randomness, we further add up to three randomly selected negative objects that are not present in the image to the final object list. This final list includes both the positive and negative objects for each image. Given this list, the VLMs are required to select objects present in the image. Since the Flickr30K dataset does not provide ground-truth object annotations, we adopt DETR [6] with a ResNet-50 [18] backbone to extract object lists from the images. The instruction datasets for each domain are named **COCO-MOS** (**M**ulti-**O**bject **S**election) and **Flickr30K-MOS**, consisting of 97K and 108K samples, respectively. Both domains adopt the 80 object categories from the COCO dataset, with identical category names and definitions.

3.3. Instruction Task 2: Context-Aware Image Caption Editing

The objective of the second task is to instruct VLMs to edit a given image caption while preserving its original context. To generate instructions, a pair of ground-truth and reference captions is required for each image. As in the first task, we adopt the images and ground-truth captions from the datasets released in the previous work [49], which consist of 97K and 108K samples from the COCO [30] and

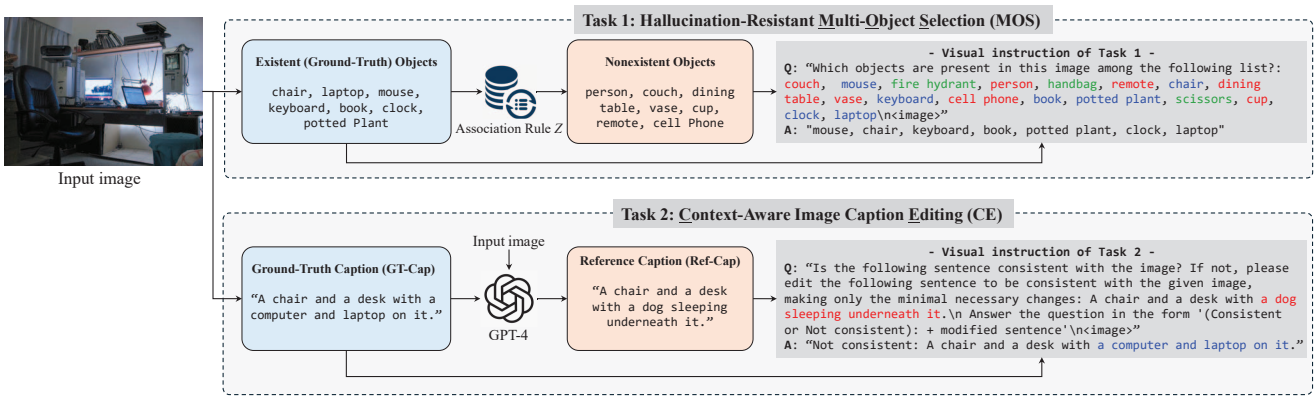


Figure 1. Overview of the proposed instruction generation framework. In Task 1, blue objects are ground-truth (existent) objects. Red and green objects are association rule-based nonexistent, and randomly selected nonexistent objects, respectively. In Task 2, the reference caption generated by GPT-4 is shown in red, while the ground-truth caption is shown in blue. Best viewed in color.

GPT-4 Prompt

System Message: You are a multimodal assistant tasked with modifying captions. Specifically, given an image and its corresponding caption, you are asked to modify the caption with the following guideline. The modified caption must include one aspect that is not consistent with the given image. The aspects are as follows:

Object existence: Modify the caption by replacing an existing objects with a non-existent one, ensuring that the changes are clearly different from the image but remain plausible.
Attribute: Misdescribe the attribute such as color, pose, position, and activity of one of the objects in the caption.
Interaction: Modify the caption to mispresent the interactions among the objects in the image.
Count: Change the caption to inaccurately represent the number of a certain object in the image while still mentioning the actual objects.

The edit distance should be smaller than 50 and greater than 5.

User Message: Based on the given image and caption, modify the caption to be inconsistent with the image based upon the given aspect. The output format should be as follows:

"image id": 279899
"GT-Cap": image caption before modification
"Ref-Cap": image caption after modification
"Type": type of aspect to generate Ref-Cap: Interaction
"Reason": the reason why the caption is inconsistent with the image

The caption (GT-Cap) is as follows: This is a cat on top of a bicycle's basket.



Input Image

GPT-4 Output

"image id": 279899,
"GT-Cap": "this is a cat on top of a bicycle's basket",
"Ref-Cap": "this is a cat playing with a dog beside a bicycle",
"Type": "Interaction",
"Reason": "The caption misrepresents the interaction by introducing a dog, which is not present in the image."

Figure 2. An example illustrating the prompt and response used to generate a reference caption using GPT-4.

Flickr30K [54] domains, respectively.

To generate reference captions, we leverage GPT4o-mini [37] (referred to as GPT4 for brevity) as a strong teacher. An example of generating a reference caption is illustrated in Fig. 2. Since the goal of image caption editing is to recognize inconsistencies in the caption and modify it, we focus on enhancing perceptual abilities. Inspired by the taxonomy of perceptual abilities introduced in [15], we define four aspects: object existence, attribute, interaction, and count as presented in the system message in Fig. 2. One of the four aspects is selected sequentially before prompting GPT-4. Given a pair of an image and its ground-truth caption, GPT4 is asked to generate a reference caption under the guidance of the selected aspect. To avoid distorting the original context of the ground-truth caption, we limit the character edit distance between the ground-truth and

generated captions to a maximum of 50. The instruction datasets for each domain are named **COCO-CE** (**C**ontext-aware caption **E**diting) and **Flickr30K-CE**, consisting of 97K and 108K samples, respectively.

Based on the generated reference captions, we establish an instruction prompt for context-aware image caption editing, as depicted in the bottom part of Fig. 1. To further enhance the analytical reasoning ability, we set half of the reference captions to be identical to the ground-truth ones, requiring VLMs to judge whether a given caption needs editing. It is worth noting that the previous methods [49, 50] always consider given captions as incorrect ones, and thus persist in editing captions. By combining the two instruction datasets, we obtain **COCO-MOS-CE** and **Flickr30K-MOS-CE** instruction datasets, consisting of 194K and 216K samples, respectively.

4. Experiments

4.1. Implementation Details

We adopt LLaVA-1.5 [34] and mPLUG-Owl2 [53] as our baseline VLMs. In all instruction tuning scenarios, the LLaVA-1.5 dataset consisting of 665K instructions is used as the base dataset for both models. For example, the model denoted as "mPLUG-Owl2 + COCO-MOS" in Table 2 is trained on the instructions from the LLaVA-1.5 and COCO-MOS datasets. Vicuna-7B [10] and Llama-2-7B [45] models are adopted as the language model backbones for LLaVA-1.5 and mPLUG-Owl2, respectively. The ViT-L/14 [13] model is adopted as the visual backbone for both VLMs. All instruction tuning experiments are conducted for a single epoch. We follow the default training settings (e.g., learning rate, batch size) from the official implementation of the two baseline models.

4.2. Evaluation on Context-Aware Caption Editing

Evaluation Settings To construct test sets for context-aware image caption editing, we adopt sample lists from the previous work [49], which consist of 5,366 and 4,910 samples in the COCO and Flickr30K domains, respectively. Given a pair of an image and its ground-truth caption, we generate a reference caption using GPT-4 as illustrated in the bottom part of Fig. 1. Based on each reference caption, we construct an input prompt in the following format: *"Please edit the following sentence to be consistent with the given image, making only the minimal necessary changes: (Ref-Cap)"*. While the training sets are balanced with both positive and negative samples, the test sets contain only negative samples, which require caption editing. For evaluation metrics, we use standard evaluation metrics, including BLEU [38], ROUGE-L [29], CIDEr [47], and SPICE [3], which are commonly employed to assess the similarity between reference and ground-truth captions. The quantitative evaluation results are demonstrated in Table 2. We report results under two evaluation scenarios: **in-domain** and **cross-domain**. In the in-domain setting, models are evaluated on the same domain as they were trained on. In contrast, the cross-domain setting refers to evaluating models on an unseen domain (e.g., trained in the COCO domain and tested in the Flickr30K domain). We provide evaluation scores using the reference captions, denoted as "Ref-Caps". Since the reference captions are generated such that the edit distance to the ground-truth captions does not exceed 50, the scores of Ref-Caps in Table 2 are relatively higher than those in Table 1. The two baseline models perform similarly or worse than Ref-Caps in both domains, indicating their limited ability to handle context-aware caption editing.

In-domain Results Adding the instructions from the explicit caption editing datasets (COCO-EE and Flickr30K-EE) [49] yields contrasting results across the two domains. While the performance degrades when using the COCO-EE instructions, incorporating Flickr30K-EE leads to better scores than the baseline models. One possible reason is that the reference and ground-truth caption pairs in Flickr30K-EE tend to exhibit clearer semantic discrepancies than those in COCO-EE. This stems from the dataset construction process in [49], where reference captions in Flickr30K-EE are selected from e-SNLI-VE [23], which provide contradiction and entailment hypothesis captions. In contrast, COCO-EE relies on ground-truth captions from negative samples that are chosen based on minimal edit distance, which often results in less distinctive semantic differences. This implies that the semantic contrast plays an important role in improving the performance of context-aware caption editing.

Adding the MOS task improves the performance of the baseline models, except for the BLEU-1 scores of LLaVA-1.5. Interestingly, the MOS task is not directly aligned with the objective of image caption editing. This suggests that enhancing the perceptual ability to recognize object existence can contribute to improved image understanding. On the other hand, the performance of models trained with MOS instructions is lower than that of models trained with explicit editing datasets (EE), which may be attributed to differences in the instruction objective. Further incorporating our proposed Context-aware caption Editing (MOS-CE) yields the best performance across both baseline models and domains. Notably, the MOS-CE setting achieves higher scores than Ref-Caps, unlike the other scenarios.

Cross-domain Results For cross-domain experiments, the models are evaluated on an unseen domain from the one they were trained on. Across both domains, the performance in the cross-domain setting is relatively lower than that of the in-domain setting due to domain discrepancy. The inclusion of explicit editing datasets (EE) leads to substantial performance degradation, indicating strong domain-specific characteristics. Adding instructions of MOS task

Table 2. Results on the proposed context-aware caption editing (CE) dataset. Metrics include BLEU-1 (B-1), BLEU-4 (B-4) [38], ROUGE-L (R) [29], CIDEr (C) [47], and SPICE (S)[3]. The best scores are shown in bold for each scenario of baseline VLMs and test domains.

COCO-CE Test set					
Model	**B-1**	**B-4**	**R**	**C**	**S**
Ref-Caps	77.3	58.2	77.9	507.6	64.7
LLaVA-1.5 [34] Baseline	75.3	56.6	77.5	511.8	65.5
mPLUG-Owl2 [53] Baseline	72.4	52.5	76.0	483.5	64.3
In-domain					
LLaVA-1.5 + COCO-EE [49]	65.7	39.1	64.4	327.2	45.6
LLaVA-1.5 + COCO-MOS	74.9	56.8	78.0	524.4	66.9
LLaVA-1.5 + COCO-MOS-CE	**89.4**	**79.9**	**89.7**	**769.8**	**83.0**
mPLUG-Owl2 + COCO-EE [49]	65.1	38.4	64.6	326.4	46.0
mPLUG-Owl2 + COCO-MOS	75.1	55.5	77.2	507.6	66.0
mPLUG-Owl2 + COCO-MOS-CE	**89.3**	**79.7**	**89.7**	**769.0**	**83.1**
Cross-domain					
LLaVA-1.5 + Flickr30K-EE [49]	59.7	46.5	69.4	441.6	59.7
LLaVA-1.5 + Flickr30K-MOS	73.6	54.3	76.7	499.9	64.9
LLaVA-1.5 + Flickr30K-MOS-CE	**88.2**	**77.9**	**88.2**	**740.5**	**81.3**
mPLUG-Owl2 + Flickr30K-EE [49]	61.7	48.2	70.0	454.5	60.7
mPLUG-Owl2 + Flickr30K-MOS	77.2	58.9	79.1	545.1	68.9
mPLUG-Owl2 + Flickr30K-MOS-CE	**87.8**	**77.0**	**87.7**	**729.4**	**80.5**

Flickr30K-CE Test set					
Model	**B-1**	**B-4**	**R**	**C**	**S**
Ref-Caps	67.3	45.8	69.6	364.8	55.5
LLaVA-1.5 [34] Baseline	50.1	30.8	61.1	287.3	47.3
mPLUG-Owl2 [53] Baseline	51.4	31.2	61.7	301.1	47.8
In-domain					
LLaVA-1.5 + Flickr30K-EE [49]	64.7	47.8	69.7	446.9	59.8
LLaVA-1.5 + Flickr30K-MOS	49.4	31.2	62.9	298.4	49.1
LLaVA-1.5 + Flickr30K-MOS-CE	**82.1**	**68.6**	**83.3**	**648.5**	**75.4**
mPLUG-Owl2 + Flickr30K-EE [49]	63.9	45.4	68.0	420.5	58.1
mPLUG-Owl2 + Flickr30K-MOS	56.8	36.5	65.3	351.7	52.3
mPLUG-Owl2 + Flickr30K-MOS-CE	**81.6**	**67.6**	**83.0**	**639.5**	**74.6**
Cross-domain					
LLaVA-1.5 + COCO-EE [49]	38.4	15.2	44.9	145.8	29.9
LLaVA-1.5 + COCO-MOS	49.3	30.2	60.8	285.9	47.4
LLaVA-1.5 + COCO-MOS-CE	**72.2**	**56.5**	**77.6**	**552.5**	**66.7**
mPLUG-Owl2 + COCO-EE [49]	42.9	18.7	49.7	175.5	32.9
mPLUG-Owl2 + COCO-MOS	57.4	36.1	65.0	337.8	51.8
mPLUG-Owl2 + COCO-MOS-CE	**73.2**	**57.4**	**78.7**	**565.1**	**67.8**

Table 3. Evaluation results on robustness against visual hallucinations using the POPE metric [28]. *Advs.*, *Popu.* and *Rand.* denote that negative objects are sampled from adversarial, popular, and random settings, as described in [28], respectively. The best scores are shown in bold and the second-best scores are underlined for each test set.

Model	**COCO** [30]						**AOKVQA** [43]						**GQA** [20]					
	Advs.		*Popu.*		*Rand.*		*Advs.*		*Popu.*		*Rand.*		*Advs.*		*Popu.*		*Rand.*	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
LLaVA-1.5 [34] Baseline	85.3	84.3	87.2	86.0	88.4	87.2	88.4	88.2	90.5	90.2	91.7	91.3	91.2	91.2	94.0	93.8	95.0	94.9
LLaVA-1.5 + COCO-EE [49]	85.5	84.4	87.1	85.9	88.2	87.0	87.6	87.3	89.7	89.3	91.0	90.4	90.9	91.0	93.7	93.5	94.7	94.6
LLaVA-1.5 + COCO-MOS	86.5	85.4	87.7	86.5	88.7	87.4	89.0	88.7	90.1	90.5	91.8	91.3	92.5	92.4	94.4	94.3	95.1	94.9
LLaVA-1.5 + COCO-MOS-CE	**86.8**	**85.8**	**88.0**	**86.8**	**89.0**	**87.8**	**89.6**	**89.3**	**91.1**	**90.7**	**92.2**	**91.8**	**92.6**	**92.6**	**94.8**	**94.7**	**95.3**	**95.2**

yields better performance than EE, highlighting the importance of perceptual capability. The further inclusion of context-aware caption editing (MOS-CE) instructions results in significant performance gains than using the EE instructions alone, demonstrating the enhanced cross-domain generalization. Overall, the results in Table 2 suggest that the proposed instruction datasets effectively enhance the ability to detect and correct inconsistencies between image-caption pairs in both in-domain and cross-domain scenarios.

Qualitative Comparison The qualitative results are presented in Fig. 3, which shows five caption editing examples from the Flickr30K-CE test set, each accompanied by the outputs of five different models. The results for TIger are obtained by adopting the released model that is trained on the Flickr30K-EE dataset [49]. The maximum number of editing rounds is set to five, following the settings described in the original paper. Compared to our instruction-based model (LLaVA-1.5 + Flickr30K-MOS-CE), TIger tends to remove words, distorting the original contexts of reference captions, as seen in (a), (b), and (c) in Fig. 3. This behavior is attributed to the deletion tagger used in TIger, which removes inconsistent words in the first round of editing. The baseline VLMs (LLaVA-1.5 and mPLUG-Owl2) often generate hallucinated captions by adhering too closely to the reference captions while failing to capture the details of image contents. Adding explicit caption editing instructions (LLaVA-1.5 + Flickr30K-EE) improves the performance of the baseline VLM, but still struggles to preserve the original contexts, as seen in (a) through (d) in Fig. 3. In contrast, the model trained on our proposed dataset successfully modifies captions while preserving the semantic context of the reference captions.

4.3. Evaluation on Visual Hallucination

To investigate the robustness against visual hallucinations, we adopt the POPE evaluation metric [28], which consists of binary questions regarding the existence of objects across the COCO [30], AOKVQA [43], and GQA [20] domains. All evaluations are conducted on test sets constructed with the 80 object categories from the COCO dataset. The evaluation results in terms of accuracy (Acc) and F1-score (F1) are presented in Table 3. Inserting EE instruction yields comparable performance on the COCO domain but results in performance degradation on the other domains. In contrast, adding our instructions (MOS and MOS-CE) improves the robustness against visual hallucination. Notably,

Input Images		Image Caption Editing Results
 (a)	Reference Caption (Input)	: A man is standing next to a horse in the street.
	Ground-Truth Caption	: A man is riding a horse in the street.
	TIger [49]	: A man is riding a horse.
	LLaVA-1.5 [34]	: A man is standing next to a horse in the street.
	mPLUG-Owl2 [53]	: A man is standing next to a horse in the street.
	LLaVA-1.5 + Flickr30K-EE [49]	: A man is riding a horse.
	LLaVA-1.5 + Flickr30K-MOS-CE (Ours)	: A man is riding a horse in the street.
 (b)	Reference Caption (Input)	: Three people dressed differently are standing near a cart and looking at vegetables.
	Ground-Truth Caption	: Two people dressed differently are standing near a cart and looking at vegetables.
	TIger	: People are pushing a cart.
	LLaVA-1.5	: Three people dressed differently are standing near a cart and looking at vegetables.
	mPLUG-Owl2	: Three people dressed differently are standing near a cart and looking at fruits.
	LLaVA-1.5 + Flickr30K-EE	: Two people are standing near a cart.
	LLaVA-1.5 + Flickr30K-MOS-CE (Ours)	: Two people dressed differently are standing near a cart and looking at vegetables.
 (c)	Reference Caption (Input)	: The man is playing fetch with a guy and his dogs.
	Ground-Truth Caption	: The man is taking a picture of a guy with his dogs.
	TIger	: The man is taking a walk with his dogs.
	LLaVA-1.5	: The man is playing fetch with a guy and his dogs in a field.
	mPLUG-Owl2	: The man is playing fetch with a guy and his dog.
	LLaVA-1.5 + Flickr30K-EE	: The man is taking pictures.
	LLaVA-1.5 + Flickr30K-MOS-CE (Ours)	: The man is taking a picture of a guy and his dogs.
 (d)	Reference Caption (Input)	: Children are pushing on the rope.
	Ground-Truth Caption	: Children are pulling on the rope.
	TIger	: Children are pulling on the rope.
	LLaVA-1.5	: Children are pushing on the rope, forming a long line.
	mPLUG-Owl2	: Children are pushing on the rope.
	LLaVA-1.5 + Flickr30K-EE	: Children are playing.
	LLaVA-1.5 + Flickr30K-MOS-CE (Ours)	: Children are pulling on the rope.

Figure 3. Qualitative comparison of caption editing results using the proposed test samples. Texts in red and blue indicate correct and incorrect descriptions, respectively. Texts in green are insertions generated by the models, which deviate from the ground-truth captions but are consistent with the given image contents. Best viewed in color.

further adding CE instructions enhances the performance of MOS, suggesting that the context-aware image caption editing task is positively correlated with robustness to visual hallucination.

5. Conclusion

In this paper, we address the task of context-aware image caption editing through visual instruction tuning with recent vision-language models (VLMs). We first demonstrate that baseline VLMs are not proficient at detecting and correcting inconsistencies between an erroneous caption and its corresponding image. To tackle this issue, we construct an instruction dataset comprising two tasks: hallucination-resistant multi-object selection (MOS) and context-aware image caption editing (CE). Experimental results on two domains using two baseline VLMs show that our proposed instruction dataset not only outperforms existing caption editing methods but also improves robustness against visual hallucinations. Further research on scaling up image caption editing to the paragraph level and providing feedback on caption revisions would be valuable directions for future work based on our study.

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